Journal of Membrane Science 320 (2008) 528–532

Contents lists available at ScienceDirect

Journal of Membrane Science

journal homepage: www.elsevier.com/locate/memsci

"Superpermeability" and "pumping" of atomic hydrogen through palladium membranes

M.L. Zheludkevich[a,b,*], A.G. Gusakov[a], A.G. Voropaev[a], E.N. Kozyrski[a], S.A. Raspopov[c], A.A. Vecher[a]

[a] Department of Chemistry, Belarus State University, 14 Leningradskaya Street, Minsk 220050, Belarus
[b] Department of Ceramics and Glass Engineering, CICECO, University of Aveiro, 3810-193 Aveiro, Portugal
[c] Department of Chemistry, University of Waterloo, 200 University Avenue West, Waterloo, Ontario, N2L 3G1, Canada

ARTICLE INFO

Article history:
Received 20 August 2007
Received in revised form 17 April 2008
Accepted 19 April 2008
Available online 26 April 2008

Keywords:
Atomic hydrogen
Permeation
Palladium membrane
Thermodynamical model
Superpermeability

ABSTRACT

Permeation of atomic as well as molecular hydrogen through palladium membranes has been investigated experimentally in the temperature range from room temperature to 200 °C and at a higher incident flux of hydrogen atoms on palladium surface than in previous studies. The results demonstrate that phenomena of 'superpermeability' and 'pumping' of atomic gases through metal membranes are of a common nature. A theoretical model based on chemical thermodynamics and diffusion theory adequately describes the quantitative relationships observed in experiments. It was found that permeability of atomic hydrogen depends strongly on the magnitude of surface incident flux and membrane temperature.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction

Palladium metal is known to dissolve substantial amounts of hydrogen at low temperatures [1]. This property makes it the best material for many applications where separation, purification or accumulation of hydrogen is required, including nuclear technology. In some technological processes hydrogen is present in atomic form as well. Therefore phenomena taking place in the atomic hydrogen/palladium system are of considerable interest both for fundamental science and for number of different applications.

An interesting phenomenon of 'superpermeation' was observed in experimental studies of atomic hydrogen transport through palladium membranes. It was found that permeation of atomic hydrogen across the membrane is several orders of magnitude faster than in case of diatomic counterpart and it can be comparable to permeability of an orifice [2,3]. These experimental studies were performed with membranes of different purity and incident flux of atomic hydrogen in the range 10^{11} atoms cm^{-2} s^{-1} to 10^{12} atoms cm^{-2} s^{-1}. Similar results were obtained for other transition metals: Fe, Ni, Nb and V [4–8]. A theoretical model has been proposed in Refs. [2,3] that describes 'superpermeation' of metal membranes by atomic hydrogen using certain assumptions. It was suggested that permeation probability of a hydrogen atom across the membrane is independent of the incident flux up to 3×10^{18} atoms cm^{-2} s^{-1} and also of temperature in the range 293–793 K. However, by other authors it was found that permeability increases by fivefold to sixfold at a higher hydrogen pressure and depends strongly on temperature [9]. Maximum permeation was observed at about 373 K. These results are contradicting to the model proposed by Lifshits in Refs. [2,3].

Another interesting phenomenon observed when atomic hydrogen interacts with permeable membranes is a 'pumping'. If the front surface of the membrane is exposed to atomic hydrogen, atoms will permeate through the membrane and recombine into molecules on the rear surface. In this way, molecular hydrogen with higher pressure can be accumulated in a closed volume behind the membrane (Fig. 1). When this system reaches chemical equilibrium, pressure of molecular hydrogen can be several orders of magnitude higher than that of atomic H at the front surface. In the case of palladium membrane, pressure ratio on the different sides of membrane (termed 'pumping coefficient') can be as high as 10^3 to 10^7 according to the data reported in Ref. [3]. Similar effect was observed for the iron, niobium and vanadium membranes too [5–7,10]. Thermodynamical interpretation of the 'pumping' phenomenon was given in [11].

* Corresponding author at: Department of Ceramics and Glass Engineering, CICECO, University of Aveiro, 3810-193 Aveiro, Portugal. Tel.: + 351 234370255; fax: + 351 234378146.
E-mail address: mzheludkevich@ua.pt (M.L. Zheludkevich).

0376-7388/$ – see front matter © 2008 Elsevier B.V. All rights reserved.
doi:10.1016/j.memsci.2008.04.045

The effects of 'superpermeation' and 'pumping' can be important for number of practical application specially in the area of thermonuclear reactors [12–14]. Superpermeable membranes can be used in different places of reactor providing different functions [14]. So, they can be used for separation of deuterium/tritium fuel from helium ash [6]. The membranes can also be installed along the walls in the pumping duct between the divertor and the cryopump or directly in the divertor region providing pumping of D/T fuel [13,14]. Membranes can be also efficient in pumping dissociated and ionized gas flowing from the extraction region of ion sources. This gas is usually difficult to pump, because of small gap between the plasma electrode and the first extraction grid. The fuel gas can be compressed, purified, and recycled back into the ion source [14]. Palladium and iron membranes were also suggested to be used for detection of atomic hydrogen near the plasma zones [15,16]. Another possible application of superpermeable membranes is recovery of hydrogen from the plasma-chemical processes used hydrogen-containing compounds. For example a multilayer membrane for recovery of hydrogen from a non-thermal plasma reactor in which the H_2S is dissociated into sulfur and hydrogen ions and atoms is patented [17].

In the present work, we investigated permeation of hydrogen through the palladium membrane exposed to the fluxes of atomic or molecular hydrogen at about 10^{18} atoms cm^{-2} s^{-1}, in the temperature range 293–523 K. Our goal was to find a quantitative relationship between 'superpermeation' and 'pumping' since these phenomena have a common nature due to a very high chemical potential of atomic hydrogen ($_{\Delta}G^{\circ}_{298}$ for dissociation of hydrogen molecule is 406.6 kJ mol^{-1}).

2. Experimental

Studies of the 'pumping' effect and permeation of hydrogen through palladium membrane were carried out at a high-vacuum setup that was used previously for experiments on the interaction of atomic oxygen beams with surfaces of metals as described elsewhere [18,19]. Vacuum chamber was evacuated down to 10^{-2} Pa before opening the hydrogen flux. Schematic of the experiment is shown in Fig. 1. Hydrogen was generated by electrolysis and then passed through a microwave discharge (2450 MHz, 60 W) in a quartz capillary in order to produce atomic hydrogen. The degree of dissociation was typically about 10%.

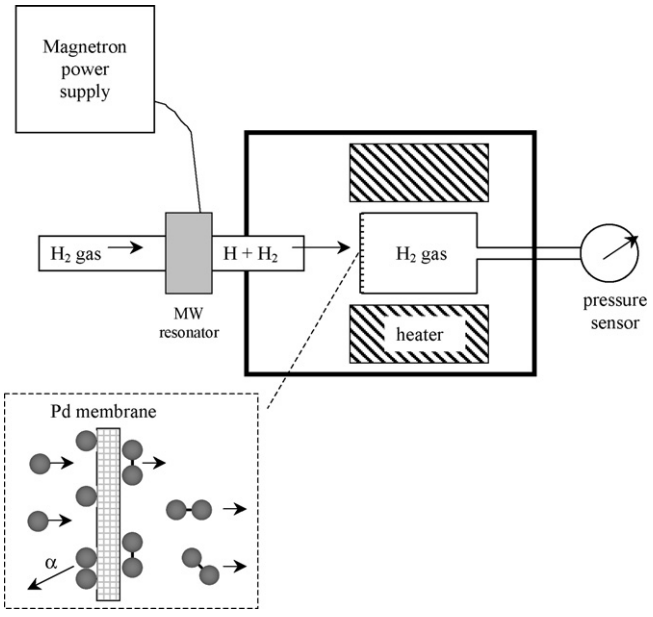


Fig. 1. Setup used for the permeation and 'pumping' experiments.

Just after passing the discharge, mixture of H and H_2 entered a high-vacuum chamber where the cell with Pd membrane and pressure sensor was located. The cell was heated to the temperature of experiment by an electrical oven. The closed cell was made by welding of Pd membrane (2 cm in diameter) to the stainless steel cage under Ar atmosphere. The steel cage has an outlet on back side for evacuation and connection of a pressure sensor. Before experiments the internal compartment of the closed cell was evacuated down to 10^{-2} Pa and kept during 72 h at atmosphere pressure. No decrease of vacuum inside the cell was revealed confirming absence of any gas leakage. Another test was conducted in helium at 10^4 Pa. Permeation of He into the internal compartment through Pd membrane was also not revealed.

Palladium foil of 99.9% purity and about 0.1-mm thick (from Alfa Aesar) was used as a membrane. Surface was washed by methanol before the experiments.

In order to experimentally confirm the effects of superpermeability (comparable to permeability of an orifice) and 'pumping', studies of the permeation of atomic and molecular hydrogen through palladium membrane have been performed. Experiments were carried out in the temperature range 293–523 K and at the fluxes of atomic and molecular hydrogen 10^{17} atoms cm^{-2} s^{-1} and 10^{18} atoms cm^{-2} s^{-1}, respectively. The pressure of atomic or molecular hydrogen in the zone of membrane was calculated using Herz-Knudsen equation [1].

3. Results

The experiments on molecular and atomic hydrogen permeability through the palladium membrane at different temperatures and different values of gas pressure were carried out using experimental setup schematically depicted in Fig. 1. Fig. 2 demonstrates increase of H_2 pressure inside the cell during exposition of the palladium membrane to atomic hydrogen at 2 Pa and different temperatures. The rate of pressure increase is strongly dependent on temperature of the membrane. It has fast increase when temperature rises from 293 K to 373 K with following decrease of permeation.

The same experiments were conducted exposing Pd membrane to molecular hydrogen using the same setup. Temperature depen-

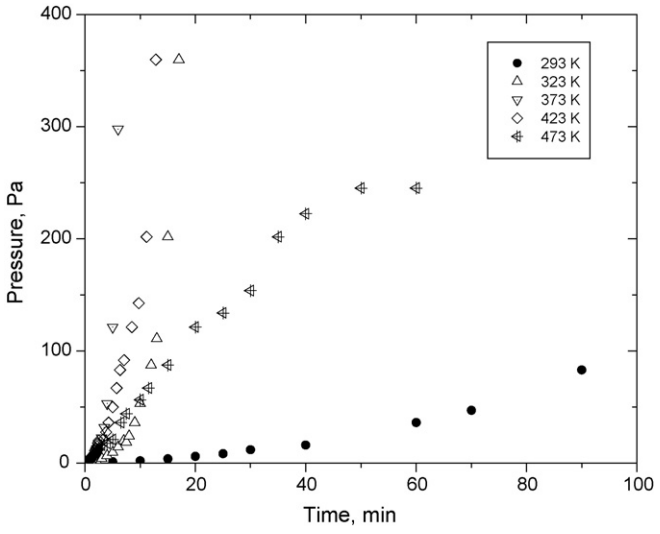


Fig. 2. Pressure of molecular hydrogen inside the cell as a function of membrane exposure time to atomic hydrogen at 2 Pa and various temperatures.

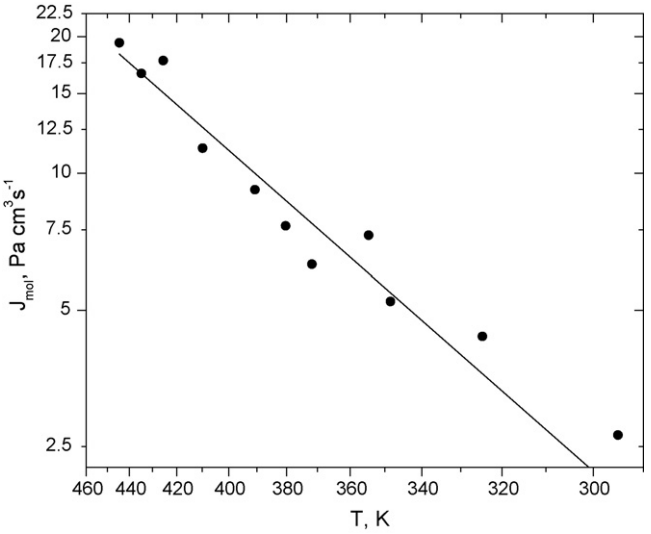


Fig. 3. Temperature dependence of the permeation rate of molecular hydrogen through palladium membrane at 10^4 Pa.

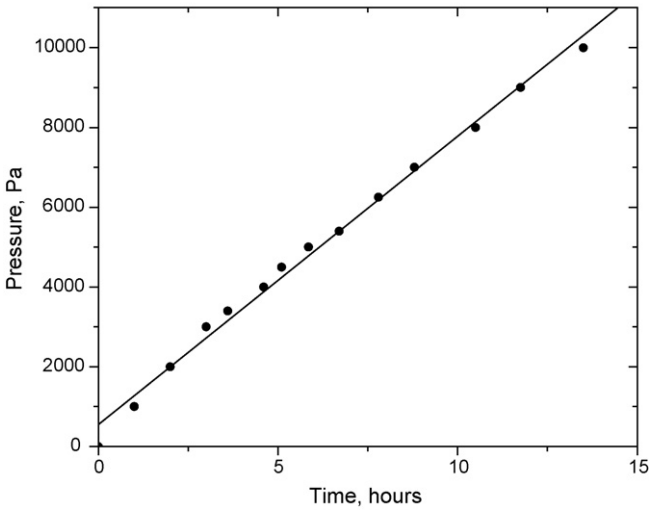


Fig. 4. Plot of the pressure of molecular hydrogen inside the cell versus time of exposure to atomic hydrogen at the membrane temperature of 363 K.

dence of the molecular hydrogen permeability shows continuous increase with temperature and can be described by classical Arrhenius law (Fig. 3). The values of permeability of molecular gas and the temperature dependence of permeability are in good agreement with data reported before [1]. Permeability of atomic hydrogen at 2 Pa and molecular hydrogen at 15 Pa and 10^4 Pa for different temperatures are summarized in Table 1. Permeability was measured after the hydrogen flux through the membrane achieves a constant value. Increase of molecular hydrogen pressure leads to sufficiently higher permeation rate in comparison to the low pressure experiments. However in the case of dissociated gas at lower temperatures permeability of atomic hydrogen at low pressure (2 Pa) is significantly higher than that of molecular one even at 10^4 Pa. These results confirm "superpermeability" of atomic hydrogen in our experiments.

A long-term experiment of hydrogen accumulation through the palladium membrane exposed to atomic hydrogen (2 Pa) was also performed at 363 K as shown in Fig. 4. Hydrogen pressure was increasing nearly linearly with time and reached 10100 Pa (76 Torr), however this experiment could not be completed because the membrane was damaged.

4. Discussion

Detailed thermodynamical analysis was performed in order to interpret the obtained results and the observed effects. Description of thermodynamic approach used in these interpretations is presented below. Molecular hydrogen permeates through metallic membrane by following main steps:

1. physical absorption of hydrogen molecules;
2. dissociation of molecules to atoms with following chemosorption;
3. transition of chemosorbed hydrogen atoms to the metal lattice;
4. diffusion of hydrogen in bulk metal;
5. transfer of hydrogen atoms to the rear surface of membrane;
6. recombination of atoms on rear surface;
7. desorption of hydrogen molecules.

When atomic hydrogen is used the first two steps can be omitted since the atoms can stick to the metal surface becoming chemisorbed without need for dissociation. This leads to sufficiently higher concentration of chemisorbed atoms on the metal surface when gas is pre-dissociated.

In the case of relatively thick metallic membranes, as one used in our experiments, the permeation of hydrogen is limited by diffusion of hydrogen atoms through the metallic membrane. As well known the driving force for hydrogen transfer through the membrane is the difference in chemical potential of chemisorbed hydrogen at two surfaces of membrane and in turn the difference in concentration of hydrogen atoms dissolved in the metal at two sides of membrane. A conventional diffusion model can be used to describe hydrogen permeation. According to Fick's law, flow of hydrogen across the membrane is given by the following equation:

$$J = D\frac{\Delta C}{\Delta x}S \quad (1)$$

where D is the diffusion coefficient of hydrogen in palladium; ΔC is the difference in concentration of dissolved hydrogen at two sides of the membrane; Δx is thickness and S is the surface area of membrane.

When the membrane is exposed to molecular hydrogen, dependence of the permeating flow on pressure can be described with

Table 1
Permeation rates of atomic and molecular hydrogen at the different values of temperature and pressure

Temperature (K)	Flow of hydrogen through the membrane (Pa cm^3 s^{-1})		
	Atomic hydrogen at 2 Pa	Molecular hydrogen at 15 Pa	Molecular hydrogen at 10^4 Pa
348	25.02	0.34	5.23
373	30.82	0.49	6.30
393	20.83	0.63	9.20
423	7.07	0.89	17.7
473	2.28	1.43	n/a

Table 2
Equilibrium values of the pressure, $P^*_{H_2}$, achieved by 'pumping' of atomic hydrogen through palladium membrane and 'pumping coefficient' calculated using Eq. (10) from experimental data obtained at 2 Pa of atomic hydrogen

Temperature (K)	Flow of atomic hydrogen (Pa cm^3 s^{-1})	$P^*_{H_2}$, Pa (Torr)	Pumping coefficient, $P^*_{H_2}/P_H$
348	25.02	64,184 (481)	32,092
373	30.82	46,622 (350)	23,311
393	20.83	12,638 (95)	6,319
423	7.07	730(5.5)	365
473	2.28	29(0.22)	15

the following equation:

$$J_{mol} = DK_S\left(\sqrt{P'_{H_2}} - \sqrt{P''_{H_2}}\right)\frac{S}{\Delta x} \quad (2)$$

where P'_{H_2} and P''_{H_2} are the pressure values of molecular hydrogen at the front and rear surfaces of membrane, respectively and K_S is the Siverts constant for solvation of hydrogen in palladium.

In the case of atomic hydrogen, Eq. (2) transforms into Eq. (3):

$$J_{at} = D\left(K_H P'_H - K_S\sqrt{P''_{H_2}}\right)\frac{S}{\Delta x} \quad (3)$$

where P'_H is the pressure (in Pa) of atomic hydrogen at the front surface of membrane and K_H is the Henry constant for solvation of atomic hydrogen in palladium.

Let us now consider the situation when the flows given by Eqs. (2) and (3) are equal. It follows from these equations that

$$K_H P'_H = K_S\sqrt{P'^*_{H_2}} \quad (4)$$

where $P^*_{H_2}$ will denote the pressure of molecular hydrogen that gives the same flow through the membrane as the atomic hydrogen at P_H [11].

We shall now express $P^*_{H_2}$ as a function of P_H using thermodynamics. In the case of chemical equilibrium between atomic and molecular hydrogen, the relationship is given by

$$P^*_{H_2} = P_H^2\exp\left(\frac{\Delta_d G^\circ}{RT}\right) \quad (5)$$

where $\Delta_d G^\circ$ is the standard Gibbs energy for dissociation of H_2 molecule.

In reality, however, a substantial fraction of hydrogen atoms will be recombining on the front surface of palladium membrane (see Fig. 1), which we shall account for by using the recombination coefficient, α:

$$P^*_{H_2} = \{(1-\alpha)P_H\}^2\exp\left(\frac{\Delta_d G^\circ}{RT}\right) \quad (6)$$

Substitution of Eqs. (4) and (6) allows to eliminate Henry constant, KH in the expression for permeation of atomic hydrogen, Eq. (3):

$$J_{at} = DK_S\left\{(1-\alpha)P_H\exp\left(\frac{\Delta_d G^\circ}{2RT}\right) - \sqrt{P''_{H_2}}\right\}\frac{S}{\Delta x} \quad (7)$$

When the membrane is exposed to atomic hydrogen, its surface is being gradually saturated with dissolved gas, therefore gradient of concentration and hydrogen flow across the membrane increases with time initially. At some point in time the concentration of hydrogen near the front surface reaches the solubility limit and flow through the membrane becomes steady. Such time dependencies of hydrogen pressure in the cell behind palladium membrane are shown in Fig. 2 for several different temperatures. As the temperature is raised from 293 K to 373 K, permeability of atomic hydrogen increases and it takes less time to reach the steady flow, because the diffusion of dissolved hydrogen becomes faster. Maximum permeability was found at 373 K. At this temperature, the probability that a hydrogen atom will penetrate across the membrane is about 3×10^{-3}. If the temperature is increased further, permeability actually decreases due to the decreasing solubility of hydrogen in palladium and higher recombination coefficient of hydrogen atoms on the surface of membrane.

In the case of molecular hydrogen, permeability increases exponentially with temperature (see Fig. 3) and decreases as a square root of pressure, in agreement with Eq. (2) as shown in Table 1.

The experimentally determined dependence of the permeation rate (in Pa cm^{-2} s^{-1}) on the temperature and pressure of molecular hydrogen has the following form:

$$J_{mol} = 7.29\times10^{-2}\exp\left(-\frac{15900}{RT}\right)\left(\sqrt{P'_{H_2}} - \sqrt{P''_{H_2}}\right)\frac{S}{\Delta x} \quad (8)$$

This equation can be simplified if P'_{H_2} is much higher than P''_{H_2} (that usually is the case when system is far from equilibrium):

$$J_{mol} = 7.29\times10^{-2}\exp\left(-\frac{15900}{RT}\right)\sqrt{P'_{H_2}}\frac{S}{\Delta x} \quad (9)$$

Comparison of the results obtained for molecular and atomic hydrogen shows that at the same conditions permeation of atomic hydrogen is much faster. This effect has been termed as 'superpermeability' of atomic hydrogen. The reason for this is that atomic hydrogen has a much chemical potential than its molecular counterpart at the same pressure and temperature. In turn the difference in chemical potential on the two sides of membrane is a driving force for hydrogen permeation. Thus use of atomic hydrogen instead of molecular one generates huge difference of chemical potentials leading to very fast permeation. Experimental values of the permeation rate through the membrane exposed to atomic or molecular hydrogen are presented in Table 1.

Another effect that has been observed in the present work is the 'pumping' of atomic hydrogen through the palladium membrane. Atomic hydrogen that has penetrated the membrane recombines on the rear surface and desorbing molecules create a pressure higher than the pressure of atomic hydrogen at the front surface (see Fig. 1). This phenomenon is also due to the high chemical potential of atomic hydrogen, which creates a higher concentration of dissolved hydrogen near the front surface of membrane. Diffusion of hydrogen through the membrane will continue until the system will reach equilibrium state where the dissolved gas concentrations at the two surfaces will become equal.

Our data on permeation rates of atomic and molecular hydrogen allow determining an upper limit (equilibrium value) for the pressure of molecular hydrogen that can be obtained in the closed volume behind a membrane. Equilibrium will be achieved when the flow caused by atomic hydrogen acting upon the front surface is equal to the oppositely directed flow created by molecular hydrogen acting upon the rear surface. So, using the experimental data on permeation rate of atomic hydrogen and the Eq. (9), one can calculate the maximum pressure of molecular hydrogen that can be achieved at certain exposure conditions to flux of atomic hydrogen

in the following way:

$$P^*_{H_2} = \left\{ \frac{J_{at}\ \Delta x}{S \times 7.29 \times 10^{-2}} \exp\left(\frac{15,900}{RT}\right) \right\}^2 \quad (10)$$

Table 2 presents the values of $P^*_{H_2}$ calculated for some experimental conditions using Eq. (10) as well as the 'pumping coefficient', a ratio of $P^*_{H_2}$ to the pressure of atomic hydrogen at the front surface of membrane. These values are decreasing with temperature, which is apparently a result of increasing recombination of H-atoms at the front surface.

A number of experiments have been also performed in order to confirm the 'pumping' effect. The maximum pressure of hydrogen accumulated in the cell (Fig. 1) at 473 K was 40 Pa (0.3 Torr) and at 423 K it was 405 Pa (3 Torr). These experimental data agree reasonably well with the calculated values of Table 2. Since the $P^*_{H_2}$ becomes much larger as the temperature decreases and diffusion becomes slower, the accumulation experiments at lower temperatures become more challenging and take much longer time.

The results presented above show that both 'superpermeation' and 'pumping' take place when atomic hydrogen interacts with palladium membrane, however the highest permeation probability observed in this work is 3×10^{-3}, which is much less that the values obtained in Refs. [2–4] at a much lower incident flux of atomic hydrogen (10^{11} atoms cm^{-2} s^{-1} to 10^{12} atoms cm^{-2} s^{-1}). So, the assumption that permeability of atomic hydrogen may remain comparable to that of an orifice up to the high incident fluxes of 10^{18} atoms cm^{-2} s^{-1} [3] does now find an experimental confirmation in this work. Another assumption that could not be confirmed is the independence of permeability on temperature. As can be seen from Fig. 2, the maximum permeability was found at 373 K. At temperatures above 523 K, dissociation of hydrogen practically does not affect the permeability, similarly to the results reported in [9]. The most likely explanation is that recombination of hydrogen atoms at the front surface of membrane becomes faster at the higher temperatures. The independence of permeability from temperature in [3,4] was probably due to the lower incident flux and surface coverage of adsorbed hydrogen atoms on palladium surface. At such conditions, increasing temperature may not cause a substantial increase in the rate of recombination.

5. Conclusions

The results of this work suggest that "superpermeability" of atomic hydrogen through the palladium membrane can be described using a diffusion model, which takes into account larger chemical potential of atomic hydrogen. Studies of the 'pumping' of atomic hydrogen through palladium membrane demonstrated that this effect has a common nature with the 'superpermeability' due to a very high chemical potential of atomic hydrogen ($_{\Delta}G^{\circ}_{298}$ for dissociation of hydrogen molecule is 406.6 kJ mol^{-1}). The 'pumping coefficient' can be calculated from experimental data on the permeation rate of atomic hydrogen using the phenomenological dependencies found in this work. A good agreement was found between the proposed model and experimental measurements of the pressure of accumulated hydrogen.

References

[1] E. Fromm, Kinetics of Metal–Gas Interactions at Low Temperatures, Springer-Verlag, Berlin, 1998.
[2] A.L. Lifshits, L.L. Metter, A.A. Samartsev, Interaction of atomic-beam of deuterium with palladium partition, J. Tech. Phys. 29 (1979) 1490.
[3] A.L. Lifshits, Superpermeability of solid membranes and gas evacuation. Part I. Theory, Vacuum 29 (1979) 103.
[4] A.I. Lifshits, M.E. Notkin, A.A. Samartsev, Physicochemical origin of superpermeability—large-scale effects of surface-chemistry on hot hydrogen permeation and absorption in metals, J. Nucl. Mater. 170 (1990) 79.
[5] H. Hackfort, K. Bösche, F. Waelbroeck, J. Winter, P. Wienhold, Hydrogen pumping and compression by superpermeation through iron, J. Nucl. Mater. 144 (1987) 10.
[6] A.I. Livshits, M.E. Notkin, V.I. Pistunovich, M. Bacal, A.O. Busnyuk, Superpermeability: critical points for applications in fusion, J. Nucl. Mater. 220–222 (1995) 259.
[7] Y. Hatano, A. Livshits, Y. Nakamura, A. Busnyuk, V. Alimov, C. Hiromi, N. Ohyabu, K. Watanabe, Influence of oxygen and carbon on performance of superpermeable membranes, Fusion Eng. Des. 81 (2006) 771.
[8] M.E. Notkin, A.I. Livshits, A.M. Bruneteau, M. Bacal, Effect of ion bombardment on plasma-driven superpermeation of hydrogen isotopes through a niobium membrane, Nucl. Instrum. Methods B 179 (2001) 373.
[9] V.V. Romanov, Yu. K. Bichtok, Investigation of hydrogen permeation through Pd alloy under gamma radiation, in: Voprosy Atomnoy Nauki I Tehniki, Ser. Atomno-Vodorodnaya Energetica I Tehnologiya (Problems of Nuclear Science and Engineering, Ser. Atomic Hydrogen Energetics and Technology) vol. 3, 1987, p. 31 (in Russian).
[10] Y. Hatano, K. Watanabe, A. Livshits, A. Busnyuk, V. Alimov, Y. Nakamura, Kenichi Hashizume, Effects of bulk impurity concentration on the reactivity of metal surface: sticking of hydrogen molecules and atoms to polycrystalline Nb containing oxygen, J. Chem. Phys. 127 (2007) 204707–204711.
[11] A.A. Vecher, Diagnostics of unequilibrium dissociated-gas, Mater. Sci. Forum 76 (1991) 195.
[12] Y. Nakamura, N. Ohyabu, H. Suzuki, Y. Nakahara, A. Livshits, M. Notkin, V. Alimov, A. Busnyuk, Development of divertor pumping system with superpermeable membrane, Fusion Eng. Des. 49–50 (2000) 899.
[13] Y. Nakamura, S. Sengoku, Y. Nakahara, N. Suzuki, H. Suzuki, N. Ohyabu, A. Busnyuk, M. Notkin, A. Livshits, Deuterium pumping experiment with superpermeable Nb membrane in JFT-2M tokamak, J. Nucl. Mater. 278 (2000) 312.
[14] M. Bacal, A.M. Bruneteau, A.I. Livshits, V.N. Alimov, M.E. Notkin, Hydrogen superpermeable membrane operation under plasma conditions, Fusion Eng. Des. 65 (2003) 423.
[15] W.T. Shmayda, J. Winter, F. Waelboreck, P. Wienhold, Radial atomic-hydrogen flux measurements to the textor liner, J. Nucl. Mater. 145–147 (1987) 201.
[16] I. Takagi, K. Toyoda, M. Katayama, H. Fujita, K. Higashi, Experiment on atomic hydrogen reflection by use of a permeation probe, J. Nucl. Mater. 258–263 (1998) 1082.
[17] C. Etievant, M. Roshd, Method and devices for producing hydrogen by plasma reformer, US Patent 6,245,309 (2001).
[18] A.G. Gusakov, A.G. Voropayev, M.L. Zheludkevich, A.A. Vecher, S.A. Raspopov, Studies of the interaction of copper surface with atomic and molecular oxygen, Phys. Chem. Chem. Phys. 1 (1999) 5311.
[19] S.A. Raspopov, A.G. Gusakov, A.G. Voropayev, A.A. Vecher, M.L. Zheludkevich, A.S. Gritsovets, V.K. Grishin, Interaction of titanium with atomic and molecular oxygen, J. Chem. Soc., Faraday Trans. 93 (1997) 2113.